

Project Design Phase

Solution Architecture

Date: 18 June 2026

Team ID: VIP-C2-Book-A-Doctor-MERN

Project Name: Book A Doctor – Online Doctor Appointment Booking System

Solution Architecture Overview

The Book A Doctor system is designed using the MERN Stack architecture. The solution consists of three major layers: Frontend Layer, Backend Layer, and Database Layer.

The architecture enables secure communication between patients, doctors, administrators, and the database through REST APIs.

Architecture Components

Frontend Layer (React.js)

The frontend provides the user interface for:

- Patient Registration and Login
- Doctor Search
- Appointment Booking
- Notifications
- Dashboard Management

Backend Layer (Node.js & Express.js)

The backend handles:

- API Requests
- Authentication & Authorization
- Appointment Processing

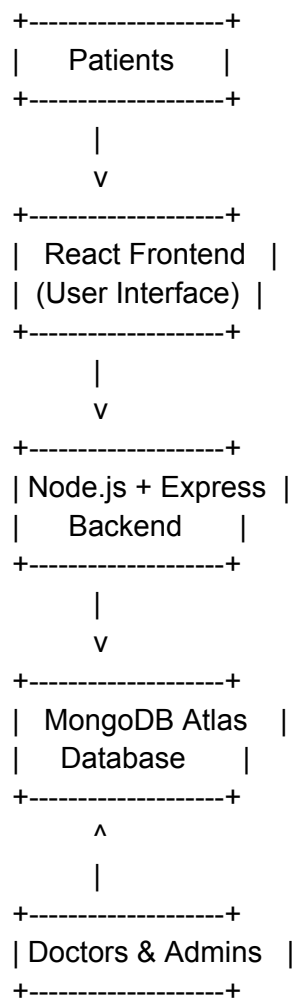
- Doctor Approval Workflow
- Notification Management

Database Layer (MongoDB Atlas)

The database stores:

- User Information
 - Doctor Information
 - Appointment Records
 - Notifications
 - System Data
-

Solution Architecture Diagram



Data Flow

1. User logs into the application.
 2. Frontend sends requests to backend APIs.
 3. Backend validates requests and processes data.
 4. MongoDB stores and retrieves application data.
 5. Results are returned to users through the frontend.
 6. Notifications are generated for appointments and approvals.
-

Benefits of the Architecture

- Scalable and modular design.
 - Secure authentication using JWT.
 - Efficient appointment management.
 - Centralized data storage.
 - Easy deployment and maintenance.
-

Conclusion

The proposed MERN Stack architecture provides a secure, scalable, and efficient solution for managing healthcare appointments. It ensures smooth communication between patients, doctors, administrators, and the database while supporting future enhancements and scalability.